Exercise Solutions: Nested For Loop

1. Right-Angle Triangle Number Pattern

Solution

```
import java.util.Scanner;

public class NumberTriangle {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input number of rows : ");
        int rows = input.nextInt();

        for (int row = 1; row <= rows; row++) {
            for (int number = 1; number <= row; number++) {
                System.out.print(number);
            }
            System.out.println();
        }
        input.close();
    }
}
```

Sample Output

```
1
12
123
1234
12345
123456
1234567
12345678
123456789
12345678910
```

2. Right-Angle Triangle Repeating Numbers

Solution

```
public class RepeatingNumberTriangle {
    public static void main(String[] args) {
        int rows = 4;

        for (int row = 1; row <= rows; row++) {
            for (int count = 1; count <= row; count++) {
                System.out.print(row);
            }
            System.out.println();
        }
    }
}
```

Sample Output

```
1
22
333
4444
```

3. Right-Angle Triangle with Incrementing Numbers

Solution

```
public class IncrementingNumberTriangle {
    public static void main(String[] args) {
        int rows = 4;
        int number = 1;

        for (int row = 1; row <= rows; row++) {
            for (int column = 1; column <= row; column++) {
                System.out.print(number + " ");
                number++;
            }
            System.out.println();
        }
    }
}
```

Sample Output

```
1
2 3
4 5 6
7 8 9 10
```

4. Pyramid with Repeating Numbers

Solution

```
public class RepeatingNumberPyramid {
    public static void main(String[] args) {
        int rows = 4;

        for (int row = 1; row <= rows; row++) {
            for (int space = 1; space <= rows - row; space++) {
                System.out.print(" ");
            }
            for (int count = 1; count <= row; count++) {
                System.out.print(row + " ");
            }
            System.out.println();
        }
    }
}
```

Sample Output

```
   1
  2 2
 3 3 3
4 4 4 4
```

5. Print Floyd's Triangle

Solution

```
import java.util.Scanner;

public class FloydsTriangle {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input number of rows : ");
        int rows = input.nextInt();
        int number = 1;

        for (int row = 1; row <= rows; row++) {
            for (int column = 1; column <= row; column++) {
                System.out.print(number + " ");
                number++;
            }
            System.out.println();
        }
        input.close();
    }
}
```

Sample Output

```
Input number of rows : 5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

6. Display Diamond Pattern

Solution

```
import java.util.Scanner;

public class DiamondPattern {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input number of rows (half of the diamond) : ");
        int rows = input.nextInt();

        for (int row = 1; row <= rows; row++) {
            for (int space = 1; space <= rows - row; space++) {
                System.out.print(" ");
            }
            for (int star = 1; star <= 2 * row - 1; star++) {
                System.out.print("*");
            }
            System.out.println();
        }
        for (int row = rows - 1; row >= 1; row--) {
            for (int space = 1; space <= rows - row; space++) {
                System.out.print(" ");
            }
            for (int star = 1; star <= 2 * row - 1; star++) {
                System.out.print("*");
            }
            System.out.println();
        }
        input.close();
    }
}
```

Sample Output

```
      *
     ***
    *****
   *******
  *********
 ***********
*************
 ***********
  *********
   *******
    *****
     ***
      *
```

7. Reverse * Triangle

Solution

```
import java.util.Scanner;

public class ReverseStarTriangle {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input the number: ");
        int rows = input.nextInt();

        for (int row = 0; row < rows; row++) {
            for (int space = 0; space < row; space++) {
                System.out.print(" ");
            }
            for (int star = 0; star < rows - row; star++) {
                System.out.print("*");
            }
            System.out.println();
        }
        input.close();
    }
}
```

Sample Output

```
******
 *****
  ****
   ***
    **
     *
```

8. Right-Angle Triangle with @ Symbols

Solution

```
import java.util.Scanner;

public class AtSymbolTriangle {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input the number: ");
        int rows = input.nextInt();

        for (int row = 1; row <= rows; row++) {
            for (int space = 1; space <= rows - row + 1; space++) {
                System.out.print(" ");
            }
            for (int symbol = 1; symbol <= row; symbol++) {
                System.out.print("@");
            }
            System.out.println();
        }
        input.close();
    }
}
```

Sample Output

```
      @
     @@
    @@@
   @@@@
  @@@@@
 @@@@@@
```